

Blank Note

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Yes — for your healthcare AI app, you should **not start by trying to collect real patient data**. Start with **public real-world healthcare datasets**, then use **synthetic patient-level data** for app development/testing. Your hackathon doc focuses on patient access, prior authorization, and revenue cycle management, so your data should support those three areas.

Best answer for your situation

For a cost-effective but high-quality app, use this strategy:

Use real public data for business insights + synthetic/de-identified data for the app workflow.

That gives you realistic analysis without exposing real patient privacy.

1. Best free/low-cost real healthcare data sources

A. CMS Data — best for cost, claims, providers, utilization, revenue insight

<https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners>

CMS should be your **first place** to look because it has public Medicare data, provider data, payment data, utilization data, hospital quality data, and pricing-related datasets.

Good for:

- Cost prediction
- Reimbursement analysis
- Provider utilization
- Claim/payment trends
- Revenue cycle research
- Healthcare business insights

Start with **Medicare Physician & Other Practitioners datasets**. They include utilization, payments, submitted charges, provider identifiers, HCPCS procedure codes, and

geography. This is very useful for analyzing what services cost, how much Medicare pays, and where utilization is high.

Example business insight:

"Which procedures have the largest gap between submitted charge and Medicare payment?"

Example model:

Predict expected reimbursement using provider type, procedure code, geography, and service volume.

B. CMS Care Compare / HCAHPS — best for patient experience

https://data.cms.gov/provider-data/dataset/dgck-sy fz?utm_source=chatgpt.com

For patient care and experience, use **HCAHPS hospital patient survey data**. It measures patient experience after inpatient hospital stays and is published through CMS. CMS says HCAHPS is a national standardized survey of hospital patients about their recent inpatient experience.

Good for:

- Patient satisfaction analysis
- Hospital comparison
- Patient experience dashboards
- Quality score prediction
- Finding hospitals with poor communication or poor discharge experience

Example business insight:

"Hospitals with lower communication scores may need better patient navigation and chatbot support."

Example app feature:

Patient experience dashboard showing communication, discharge, cleanliness, and overall rating.

C. AHRQ HCUP / HCUPnet — best for hospital utilization and ED/inpatient trends

https://www.ahrq.gov/data/hcup/index.html?utm_source=chatgpt.com

AHRQ's HCUP tools let you generate healthcare statistics for inpatient and emergency department settings. HCUPnet provides national and regional hospital statistics, trends, tables, and graphs.

Good for:

- Hospital demand forecasting
- ED utilization
- Length-of-stay analysis
- Disease/service trends
- Population-level healthcare utilization

Example business insight:

"Which diagnosis groups are driving the highest emergency department utilization?"

Example app feature:

Dashboard showing service demand by region, age group, or condition.

D. NPPES / NPI Registry — best for provider directory and access

https://npiregistry.cms.hhs.gov/search?utm_source=chatgpt.com

NPPES gives public National Provider Identifier data. The NPI Registry is a free directory of active NPI records, and CMS also provides downloadable NPI files.

Good for:

- Provider search
- Provider directory
- Specialty lookup
- Location analysis
- Referral routing
- Network adequacy analysis

Example business insight:

"Which ZIP codes have fewer providers for a specialty?"

Example app feature:

Patient searches for available providers by location, specialty, and insurance

network.

E. CMS Open Payments — best for provider/industry relationship analysis

https://www.cms.gov/priorities/key-initiatives/open-payments?utm_source=chatgpt.com

Open Payments is a public CMS database showing payments and transfers of value from drug/device companies to physicians and teaching hospitals. CMS describes it as a national disclosure program for transparency in healthcare.

Good for:

- Provider transparency research
- Conflict-of-interest analysis
- Physician profile enrichment
- Healthcare business intelligence

Example business insight:

"Which specialties receive the highest industry payments by state?"

This is less important for your current patient access/prior auth/revenue cycle app, but it can be useful for broader healthcare analytics.

F. CMS Medicare Coverage Database / Coverage API — best for prior authorization logic

https://api.coverage.cms.gov/?utm_source=chatgpt.com

For prior authorization, the hardest part is finding real structured data. A good starting point is CMS coverage data, including National Coverage Determinations and Local Coverage Determinations. CMS has a Coverage API that provides access to Medicare coverage data from the Medicare Coverage Database.

Good for:

- Coverage rule lookup
- Medical necessity rules
- Prior auth support
- Documentation checklist generation
- RAG knowledge base for payer policy reasoning

Example business insight:

"Which services require strict documentation before approval?"

Example app feature:

AI reads a requested procedure and checks coverage rules to suggest missing documents.

G. Transparency in Coverage machine-readable files — best for pricing data

https://www.cms.gov/priorities/healthplan-price-transparency/overview/use-pricing-information-published-under-transparency-coverage-final-rule?utm_source=chatgpt.com

For price transparency, payers publish machine-readable files with negotiated rates and allowed amounts. CMS says plans and issuers make these files available on their websites; CMS itself does not actively publish all files centrally.

Good for:

- Cost estimate tools
- Price comparison
- Network pricing analysis
- Procedure cost transparency
- Business pricing insight

Warning: these files can be **huge and difficult to process**. They are powerful but not beginner-friendly.

Example business insight:

"How much does the same CPT code vary across payers or provider networks?"

Example app feature:

Upfront patient cost estimate before appointment.

H. MIMIC-IV — best for real clinical EHR research

https://physionet.org/about/faqs/?utm_source=chatgpt.com

MIMIC-IV is a real clinical dataset used heavily in research. It is available through PhysioNet, but it requires credentialed access. PhysioNet says users must obtain

individual access rights for credentialed datasets such as MIMIC and eICU, and sharing access is not permitted.

Good for:

- Clinical prediction
- Readmission prediction
- Length of stay prediction
- ICU research
- Clinical notes/NLP
- Risk modeling

Use this if you want more serious clinical ML research. But for your current app, MIMIC may be more complex than necessary.

2. Best synthetic data sources for app development

Even though you asked for real data, for building the actual app safely, use synthetic data.

https://ecqi.healthit.gov/tool/synthea%E2%84%A2?utm_source=chatgpt.com

A. Synthea

Synthea generates realistic but not real patient records. HealthIT.gov describes it as an open-source synthetic patient generator that creates realistic patient data free from cost, privacy, and security restrictions.

Use it for:

- Patient records
- Encounters
- Conditions
- Medications
- FHIR data
- Demo app workflows
- Chatbot testing
- RAG testing

This is probably the **best dataset for your prototype**.

B. CMS SynPUF

https://www.cms.gov/data-research/statistics-trends-and-reports/medicare-claims-synthetic-public-use-files?utm_source=chatgpt.com

CMS Medicare Synthetic Public Use Files are synthetic Medicare claims data. CMS says they were created to help users gain familiarity with Medicare claims data while protecting beneficiary privacy.

Use it for:

- Claims workflow
- Revenue cycle demo
- Claim status simulation
- Payment modeling
- Denial-risk prototype

C. CMS Blue Button 2.0 Sandbox

https://bluebutton.cms.gov/api-documentation/?utm_source=chatgpt.com

CMS Blue Button has a developer sandbox with synthetic Medicare enrollee data and the same endpoints/resource types/parameters as production.

Use it for:

- API integration practice
- FHIR-style Medicare claims workflow
- Patient-facing app testing
- OAuth/API workflow practice

This is very useful if you want your app to feel like a real health data app.

3. No-show prediction data

For patient access, no-show prediction is a great first ML feature. There are public datasets for appointment no-shows, including a Mendeley dataset from a rehabilitation center in Brazil and Kaggle datasets with around 100k appointments. The Mendeley dataset states it was created for studying medical appointment no-shows and came from an outpatient rehabilitation clinic in Brazil.

<https://data.mendeley.com/datasets/wm6w2fvkfj/1>

Use it for:

- No-show prediction
- Scheduling optimization
- Reminder strategy

- Patient access analytics

Good model ideas:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

Target variable:

- Show / no-show

Potential features:

- Age
- Gender
- Appointment day
- Scheduled day
- Waiting time
- SMS reminder
- Neighborhood
- Medical condition flags
- Previous no-show history, if available

Business insight:

"Patients with longer wait times and no reminder history are more likely to miss appointments."

App feature:

Staff dashboard showing high-risk appointments and suggested reminder actions.

4. What datasets match your app modules?

App module	Best data source	What you can build
Patient Access	No-show dataset, NPPES, HCAHPS	Scheduling dashboard, provider search, no-show prediction
Insurance Verification	CMS Blue Button Sandbox, synthetic data	Eligibility workflow simulation
Cost Transparency	CMS payment data, Transparency in Coverage files	Cost estimate and reimbursement analysis
Prior Authorization	CMS Coverage API, MCD, payer policy documents	Prior auth rule lookup, documentation checklist, RAG
Revenue Cycle	CMS SynPUF, CMS payment/utilization data	Claim review, denial risk prototype, payment dashboard
Patient Experience	HCAHPS, Care Compare	Patient satisfaction dashboard
Clinical Research	MIMIC-IV	Readmission, length of stay, clinical risk models

5. What business insights you can produce

For your app, I would focus on these insights:

Insight 1: No-show risk

Question:

Which patients are most likely to miss appointments?

Business value:

- Reduce empty appointment slots
- Improve access
- Reduce revenue loss
- Improve continuity of care

Possible output:

"High-risk patients should receive 2 reminders: one 48 hours before and one same-day."

Insight 2: Cost transparency

Question:

How much does a service usually cost, and how much does insurance usually reimburse?

Business value:

- Better patient trust
- Fewer billing surprises
- Improved payment collection
- Better patient experience

Possible output:

"Show estimated patient responsibility before the visit."

Insight 3: Prior authorization missing-document risk

Question:

Which authorization requests are likely to be delayed because documentation is incomplete?

Business value:

- Faster approvals
- Fewer delays in care
- Lower staff workload
- Better patient satisfaction

Possible output:

"This case is missing diagnosis code, previous therapy history, and clinical note summary."

Insight 4: Claim denial risk

Question:

Which claims are most likely to be denied before submission?

Business value:

- Reduce denial rate

- Improve cash flow
- Reduce rework
- Improve revenue cycle performance

Possible output:

"This claim has high denial risk because documentation does not support the procedure code."

Insight 5: Provider access gap

Question:

Which areas have fewer providers or longer access delays?

Business value:

- Better scheduling strategy
- Referral optimization
- Better network planning

Possible output:

"ZIP codes with fewer specialists may need telehealth routing or referral partnerships."

Phase 1: Build the app using free/synthetic data

Use:

- Synthea for patient records
- CMS SynPUF for claims
- CMS Blue Button Sandbox for API-style workflow
- NPPES for provider directory
- HCAHPS for patient experience
- CMS payment/utilization data for revenue insights

This keeps cost low and avoids privacy problems.

Phase 2: Build real business analytics

Use:

- CMS payment/utilization data
- HCAHPS
- AHRQ HCUPnet
- NPES
- Open Payments
- CMS Coverage API
- Public payer policy documents

This lets you create dashboards and insights without needing private patient data.

Phase 3: Add ML models

Start with:

1. **No-show prediction**
2. **Claim denial risk scoring**
3. **Cost/reimbursement prediction**
4. **Prior auth missing-document classifier**
5. **Patient experience score analysis**

For your first version, use simple models:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

Do not start with deep learning. It is usually unnecessary and more expensive.

Phase 4: Add RAG chatbot

Use RAG for documents such as:

- Patient FAQs
- Prior authorization rules
- CMS coverage documents
- Billing policies
- Insurance policy documents
- Internal workflow guides

Your chatbot should answer questions like:

- "Does this service need prior authorization?"
- "What document is missing?"

- "Why is this claim high risk?"
- "What is the estimated cost?"
- "Which patients need reminders today?"

8. Best database/app stack to stay cost-effective

For your current stage, I recommend:

Layer	Cost-effective choice
Frontend	Next.js + Tailwind CSS
Backend	FastAPI or Node.js/Express
Database	PostgreSQL
Analytics	Python, Pandas, DuckDB
Dashboard charts	Recharts or Plotly
ML	scikit-learn, XGBoost, LightGBM
RAG	LangChain or LlamaIndex
Vector DB	pgvector, Chroma, or FAISS
Hosting	Vercel for frontend, Render/Fly.io/Railway for backend, Supabase/Neon for Postgres
File storage	Supabase Storage or S3-compatible storage

For the cheapest high-quality MVP, I would use:

Next.js + FastAPI + PostgreSQL + pgvector + Python ML + CMS/Synthea/SynPUF data

10. Best starting datasets list

Start with these:

1. **CMS Medicare Physician & Other Practitioners**
2. For utilization, charges, payments, procedure-level insights.
3. **CMS HCAHPS / Care Compare**
4. For patient experience and hospital quality.
5. **AHRQ HCUPnet**
6. For hospital utilization, ED, inpatient, and population-level statistics.
7. **NPPES / NPI Registry**

8. For provider directory and specialty/location search.
9. **CMS Coverage API**
10. For Medicare coverage rules and prior authorization-style logic.
11. **CMS SynPUF**
12. For synthetic claims workflow and revenue cycle app development.
13. **Synthea**
14. For synthetic patient records and FHIR-style app testing.
15. **MIMIC-IV**
16. For advanced clinical research after credentialed access.